East West University
Department of Computer Science and Engineering
Course: CSE302
Instructor: Antu Chowdhury

LAB 02: Working with Aggregate Functions and Pattern Matching in SQL

Lab Objectives:

• Understand the concept and purpose of aggregate functions in SQL.
• Apply aggregate functions such as COUNT(), SUM(), AVG(), MAX(), and MIN() in queries.
• Understand the use of the LIKE operator for pattern matching in textual data.
• Practice writing queries using wildcard characters (% and _) with LIKE to filter string values.
• Develop the ability to combine filtering and aggregation in meaningful query patterns.

Topics Covered:

Aggregate Functions:
• Definition and purpose of aggregate functions
• Use of common functions: COUNT(), SUM(), AVG(), MAX(), MIN()
• Syntax and examples of each function
• Using GROUP BY to apply aggregate functions by group
• Handling NULL values in aggregate functions

Pattern Matching with LIKE Operator:
• Use of LIKE in WHERE clause for partial string matching
• Wildcard characters:
– % (zero or more characters)
– _ (exactly one character)
• Practical examples using names, departments, etc.

Lab Outcomes:

1. Use SQL aggregate functions to summarize data across rows.
2. Apply the COUNT() function to count total, non-null, or distinct values.
3. Use SUM() and AVG() to calculate totals and averages from numeric columns.
4. Retrieve maximum and minimum values using MAX() and MIN().
5. Use GROUP BY to apply aggregate functions per group (e.g., by department).
6. Write SQL queries using the LIKE operator with % and _ for flexible pattern matching.
7. Filter textual data using conditions like:
   – Names starting with specific letters
   – Fields that match exact character lengths
   – Strings containing or ending with certain patterns
8. Combine WHERE and LIKE with other SQL clauses to build more complex queries.

Lab Activities:

1. Table Creation and Data Insertion

(a) Write a SQL statement to create a table `Employee` with the following attributes:

- `E_id` (INT)
- `E_name` (VARCHAR)
- `dept` (VARCHAR)
- `salary` (INT)

(b) Write SQL statements to insert the following records into the `Employee` table:

E_id	E_name	Dept	Salary
1	Robin	HR	10000
2	Akash	Marketing	20000
3	Robi	HR	30000
4	Noyon	IT	40000
5	Fahim	QA	40000
6	Rakib	IT	NULL

2. Aggregate Function:

An aggregate function is a function that performs a calculation on a set of values, and returns a single value. Aggregate functions are often used with the `GROUP BY` clause of the `SELECT` statement. The `GROUP BY` clause splits the result-set into groups of values and the aggregate function can be used to return a single value for each group.

The most commonly used SQL aggregate functions are:

- `MIN()` - returns the smallest value within the selected column
- `MAX()` - returns the largest value within the selected column
- `COUNT()` - returns the number of rows in a set
- `SUM()` - returns the total sum of a numerical column
- `AVG()` - returns the average value of a numerical column

Aggregate functions ignore null values (except for `COUNT(*)`).

A. COUNT() Function

Used to count the number of rows or non-null values in a column.

📌 **Syntax:**

```
SELECT COUNT(*) FROM table_name;
SELECT COUNT(column_name) FROM table_name;
```

📌 **Example Questions:**

- Write a query to count the total of tuples in the table
- Write a query to count how many employees have a salary value.
- Write a query to counts the number of unique salaries in the Employee table.
- Write a query to count the number of employees in each department.

B. SUM() Function

Used to calculate the total sum of a numeric column.

📌 **Syntax:**

```
SELECT SUM(column_name) FROM table_name;
```

📌 **Example Questions:**

- Write a query to find the total salary of all employees.
- Write a query to calculate the total salary of employees in the HR department.
- Write a query to find the sum of salaries by department.

C. AVG() Function

Used to calculate the average value of a numeric column.

📌 **Syntax:**

```
SELECT AVG(column_name) FROM table_name;
```

📌 **Example Questions:**

- Write a query to find the average salary of all employees.
- Write a query to calculate the average salary in the QA department.
- Write a query to find the average salary in each department.

D. MAX() Function

Returns the highest value in a column.

📌 **Syntax:**

```
SELECT MAX(column_name) FROM table_name;
```

📌 **Example Questions:**

- Write a query to find the highest salary among employees.
- Write a query to find the maximum salary in the IT department.
- Write a query to display the highest salary for each department.

E. MIN() Function

Returns the lowest value in a column.

📌 **Syntax:**

```
SELECT MIN(column_name) FROM table_name;
```

📌 **Example Questions:**

- Write a query to find the lowest salary among all employees.
- Write a query to find the minimum salary in the HR department.
- Write a query to display the lowest salary for each department.

3. LIKE Operator and Wildcard Characters

The `LIKE` operator is used in a `WHERE` clause to search for a specified pattern in a column. It is often combined with wildcard characters to filter rows based on partial matches.

Common Wildcard Characters:

- % --Represents zero or more characters.
- _ --Represents a single character.

A. Basic LIKE Syntax

```
SELECT column1, column2, ...
FROM table_name
WHERE column_name LIKE 'pattern';
```

B. Examples of Wildcard Usage

Wildcard Pattern	Description	Example Match
'R%'	Starts with 'R'	Robin, Robi, Rakib
'%k%'	Contains 'k' anywhere	Akash, Rakib
'____n'	Exactly 5 characters, ends with 'n'	Robin, Noyon
'___'	Exactly 3 characters	Robi (no), Fahim (no), Akash(no)

C. Example Questions:

- Write a query to find all employees whose name ends with the letter 'i'.
- Write a query to find employees whose department name starts with 'M'.
- Write a query to find employees whose name contains the letter 'o' as the second character.
- Write a query to find employees whose department name does not contain the letter 'a'.
- Write a query to find employees whose names start with 'R' and have exactly 5 characters.
- Write a query to find employees whose salary is not NULL and whose name contains 'n'.
- Write a query to find employees whose name contains at least one 'i' and ends with 'n'.
- Write a query to find employees whose department name ends with the letter 'T'.
- Write a query to find employees whose name contains exactly one 'a'.

Problem Practice:

Lab Task # 01 (Create table):

1.(a). Write SQL statement to create a table 'instructor_your_student_id' which has 4 attributes –

i) id (int)
ii) name (varchar)
iii) dept_name (varchar)
iv) salary (decimal)
Example: create table instructor_2020360001 (……);

1.(b). Write SQL statement to create a table 'course_your_student_id' which has 4 attributes –

i) course_id (varchar)
ii) title (varchar)
iii) dept_name (varchar)
iv) credits (int)
Example: create table course_2020360001 (……);

Lab Task # 02 (Inserting data into a table):

2.(a). Write SQL statements to insert the following records into 'instructor_your_student_id' table:

ID	Name	Dept_Name	Salary
10101	Srinivasan	Comp. Sci.	65000.50
12121	Wasif	Finance	90000
15151	Mozart	Music	40000
22222	Einstein	Physics	95000
32343	El Said	History	60000
33456	Goblin	Physics	87000.50
45565	Katz	Comp. Sci.	75000
58583	Califieri	History	62000

2.(b). Write SQL statements to insert the following records into 'course_your_student_id' table:

course_id	title	dept_name	credits
BIO-101	Intro. to Biology	Biology	4
BIO-301	Genetics	Biology	4
BIO-399	Computational Biology	Biology	3
CS-101	Intro. to Computer Science	Comp. Sci.	4
CS-190	Game Design	Comp. Sci.	4
CS-319	Image Processing	Comp. Sci.	3
CS-347	Database System Concepts	Comp. Sci.	3
EE-181	Intro. to Digital Systems	Elec. Eng.	3
FIN-201	Investment Banking	Finance	3

Lab Task #03: Writing SQL Queries

1. Write a query to count the total number of instructors.
2. Write a query to find the average salary of instructors in the "Comp. Sci." department.
3. Write a query to find the highest salary among all instructors.
4. Write a query to list each department with the total salary paid to instructors in that department.
5. Write a query to count how many instructors work in each department.
6. Write a query to find the minimum salary among instructors in the "Physics" department.
7. Write a query to find all instructors whose names start with the letter 'S'.
8. Write a query to find courses whose titles start with "Intro".
9. Write a query to find all courses in the "Comp. Sci." department with credits equal to 4.
10. Write a query to count how many courses each department offers.
11. Write a query to find all instructors whose names contain the letter 'a'.
12. Write a query to find all courses whose title contains the word 'Biology' anywhere.
13. Write a query to find instructors whose department name ends with 'ics'.
14. Write a query to find courses whose course_id starts with 'CS'.
15. Write a query to find instructors whose names have exactly 6 characters.

Submission Instructions:

1. For each and every question:
   - Take **screenshots** of the **SQL query execution** and **output/result** in XAMPP or your preferred database environment.
   - Insert both the **SQL solution** and the corresponding **output image** into a **Word document (.doc/.docx)**.
2. **Save the document as a PDF** using the following file naming format: 2022-1-60-001_LAB02.pdf
   (Replace the ID with your own student ID.)
3. **Submit** the PDF file using the designated assignment section provided in the Classroom.